

Total No. of Printed Pages: 03

SUBJECT CODE NO: RNP-8013
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A) (NEP) Sem - I
Examination May/June-2026
Engineering Mechanics

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

1. Question No.1 is compulsory.
2. Solve any 4 questions from remaining questions.
3. Use of non-programmable Calculator is allowed.

Q1 Select the correct option.**20**

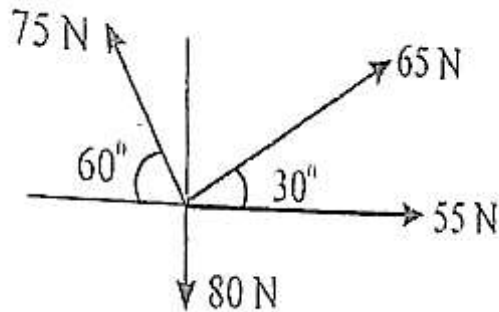
1. The centroid of a triangle lies at:
a) Midpoint of base
b) Intersection of medians
c) Intersection of altitudes
d) Intersection of diagonals
2. Centre of gravity is the point where:
a) Area is maximum
b) Volume is minimum
c) Entire weight acts
d) Force is zero
3. Moment of inertia depends on:
a) Area distribution
b) Density only
c) Temperature
d) Pressure
4. Radius of gyration is defined as:
a) Distance from centroid
b) Distance where entire area is assumed concentrated
c) Length of body
d) Width of body
5. Motion along a straight line is called:
a) Curvilinear motion
b) Circular motion
c) Rectilinear motion
d) Angular motion
6. Acceleration due to gravity is approximately:
a) 5 m/s^2
b) 9.81 m/s^2
c) 15 m/s^2
d) 20 m/s^2
7. Projectile motion follows:
a) Circular path
b) Straight line
c) Parabolic path
d) Elliptical path
8. Newton's second law states:
a) $F = ma$
b) $F = mv$
c) $F = m/a$
d) $F = ma^2$
9. Impulse is equal to:
a) Force $\times$ distance
b) Force $\times$ time
c) Mass $\times$ velocity
d) Work done

10. Law of conservation of momentum states:

- a) Momentum increases b) Momentum decreases
c) Total momentum remains constant d) Momentum becomes zero

Q.2. a) Find magnitude and direction of given force system.

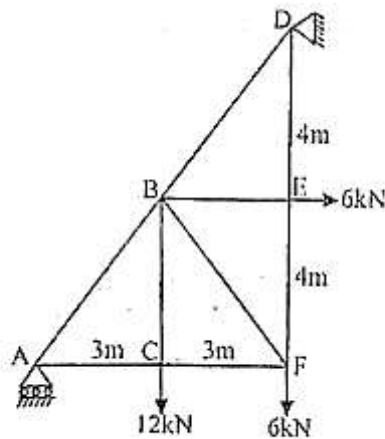
10



b) Explain in details: Lami's Theorem.

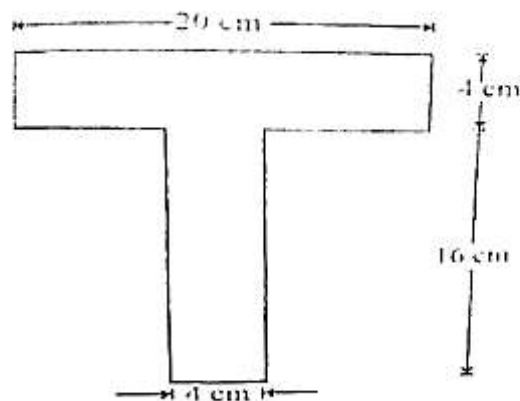
Q.3. Analyze the given truss by joint method.

10



Q.4. Find the moment of inertia about the X & Y axis.

10



Q.5. Displacement of a particle is given by: $S = 30t + 12t^2 - 2t^3$

10

Find:

- a) Velocity and acceleration at
- b) Time when velocity is maximum
- c) Maximum velocity of the particle

Q.6. A ball is thrown with a velocity of 50 m/s at an angle of elevation of 40° with the horizontal.

10

Find:

- a) Maximum height reached by the ball
- b) Time of flight
- c) Range of ball

Q.7. A stone is dropped from the top of a tower. After 3 seconds another stone is thrown downwards from the top of the same tower with an initial velocity of 35 m/s . If both the stones reach the ground at the same time determine-

10

- i. The time taken by the first stone to reach the ground
- ii. Height of the tower